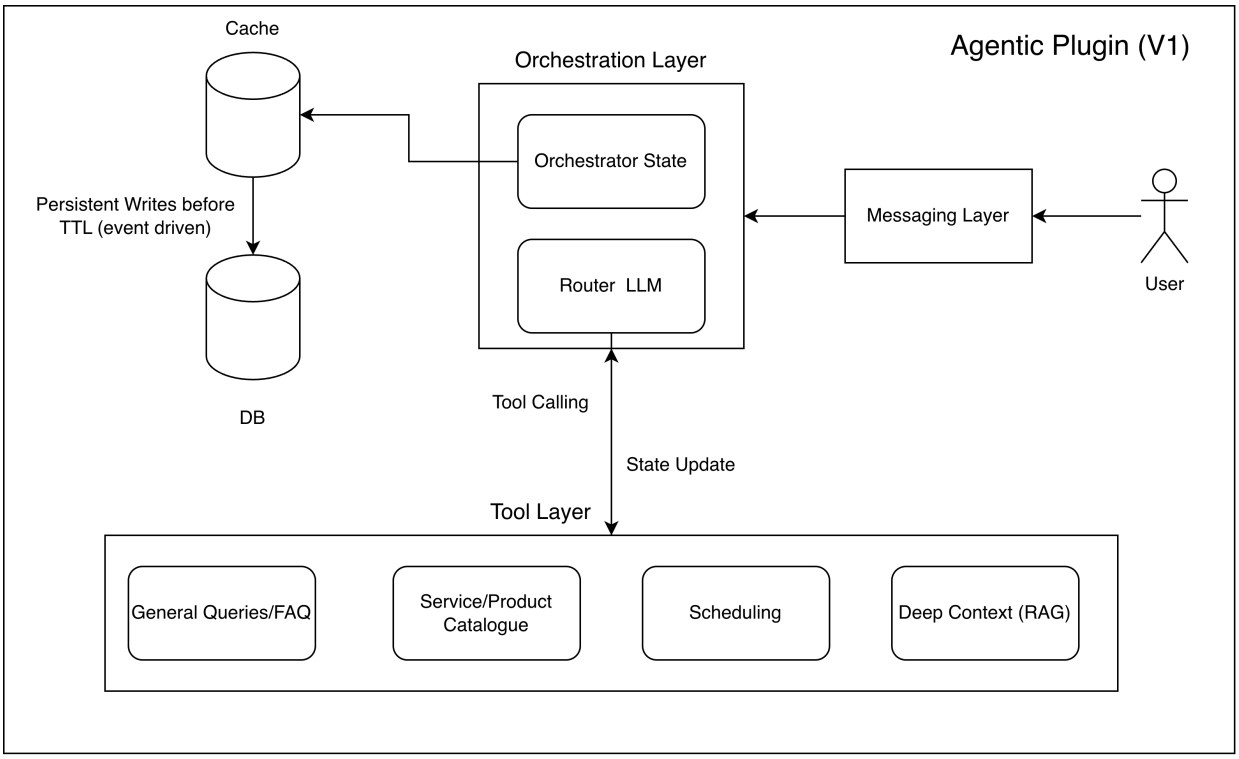
[Architecture] High Level System Design



Orchestrator State

```
{
  "conversation_id: null
  "active_agent": null,
  "flow_step": null,
  "flow_locked": false,

  "entities": {},

  "last_intent": null,
}
```

Router

```
{
  "intent": null,
  "confidence": null,
  "agent": null
}
```

Example Flow :

Scenario 1 :

1. **User :** I would like to get a haircut? what is the price and service you offer?

```
{
"intent":service_query,
"confidence":0.98,
"agent":service_catalogue

}
```

Orchestrator state update :

```
{
"conversation_id":123,
"active_agent":service_catalogue,
"active_flow":null,
"flow_step":null,
"flow_locked":false,

"entities":
{
"service_request":haircut
},

"last_intent":null

}
```

```
graph TD
  router --> service_catalogue
```

Bot : We have haircuts for Rs. 300, 500, 700 depending on the stylist.

2. **User :** What is the difference between the stylists?

```
{
"intent":general_query,
"confidence":0.98,
"agent":FAQ
}
```

Orchestrator state update :

```
{
"conversation_id":123,
"active_agent":faq,
"flow_step":null,
"flow_locked":false,

"entities":
{
"service_request":haircut
},

"last_intent":service_query

}
```

```
graph TD
  router --> faq
```

Bot : The stylist expertise increases with increasing price and the Rs. 500 and 700 haricuts also include a hairwash with 700 using suphate free premium products.

3. **User :** I would like to make an appointment for tomorrow evening at 5PM.

```
{
"intent":appointment,
```

```
"confidence":0.98
"agent": scheduling
}
```

Orchestrator state update :

```
{
"conversation_id":123,
"active_agent":scheduling,
"active_flow":scheduling,
"flow_step":stylist,
"flow_locked":true,

"entities":
{
"service_request":haircut
"time":5PM
"date":23rd march
},

"must_have":[service_request, time, date, name, mobile, styli
st],

"last_intent":faq
}
```

```
graph TD
  router --> scheduling
```

Bot : Sure, I'll schedule an appointment for 5PM tomorrow, Can you please specify your name, mobile number and the stylist you want?

4. **User : -** *Responds* -

Router has access to the orchestration state and the response - Semantically the intent will remain the same and the response is routed to the scheduling agent

```
{
"intent":appointment,
"confidence":0.98,
"agent": scheduling,
}
```

Orchestrator state update :

```
{
"conversation_id":123,
"active_agent":scheduling,
"active_flow":scheduling,
"flow_step":null,
"flow_locked":true,

"entities":
{
"service_request":haircut,
"time":5PM,
"date":23rd march,
"name":harish,
"mobile":9035790945,
"stylist":expert,
},

"must_have":[service_request, time, date, name, mobile, styli
st],
```

```
"last_intent":faq
}
```

```
graph TD
  router --> scheduling
```

- Appointment is scheduled -

Scenario 2 (continuing from step 3):

1. **User :** I would like to make an appointment for tomorrow evening at 5PM.

```
{
"intent":appointment,
"confidence":0.98,
"agent": scheduling,
}
```

Orchestrator state update :

```
{
"conversation_id":123,
"active_agent":scheduling,
"active_flow":scheduling,
"flow_step":stylist
"flow_locked":true

"entities":
{
"service_request":haircut,
"time":5PM,
"date":23rd march,
```

```
},

"must_have":[service_request, time, date, name, mobile, styli
st]

"last_intent":faq
}
```

```
graph TD
  router --> scheduling
```

Bot : Sure, I'll schedule an appointment for 5PM tomorrow, Can you please specify your name, mobile number and the stylist you want?

4. **User : -** *Responds* - but before that can you also tell me how much pedicure costs?

Multiple tool invoke : semantically service_catalogue comes before appointment. So intent switches to service_catalogue and router directly updates the entities in orchestrator state maintaining the active_flow as scheduling

```
{
"intent":service_query,
"confidence":0.98,
"agent": service_catalogue,
}
```

Orchestrator state update :

```
{
"conversation_id":123,
"active_agent":scheduling,
```

```
"flow_step":null
"flow_locked":true

"entities":
{
"service_request":haircut
"time":5PM
"date":23rd march
"name":harish
"mobile":9035790945
"stylist":expert
},

"must_have":[service_request, time, date, name, mobile, styli
st]

"last_intent":faq
}
```

```
graph TD
  router --> service_catalogue
```

Since the active flow is scheduling, the bot will suffix with a continue message

Bot : Pedicure costs Rs. 850, would you like to continue scheduling an appointment?

User : yes, but change the timing to 7PM

```
{
"intent":appointment,
"confidence":0.98,
```

```
"agent": scheduling,
}
```

Orchestrator state update :

```
{
"conversation_id":123,
"active_agent":scheduling,
"flow_step":null
"flow_locked":true

"entities":
{
"service_request":haircut
"time":7PM
"date":23rd march
"name":harish
"mobile":9035790945
"stylist":expert
},

"must_have":[service_request, time, date, name, mobile, styli
st]

"last_intent":faq
}
```

- Appointment is scheduled -